

Question	Marking Guidance	Mark	Comments						
08.1	One mark for each row. If values do not match the given unit, max 1 .	2 (2 x AO3)	Accept dm ³ / mm ³ for volume unit.						
	<table><tr><td>Concentration of copper sulfate solution / g kg⁻¹</td><td>Volume of 100 g kg⁻¹ copper sulfate solution / cm³</td><td>Volume of water / cm³</td></tr><tr><td>75</td><td>22.5</td><td>7.5</td></tr></table>		Concentration of copper sulfate solution / g kg⁻¹	Volume of 100 g kg⁻¹ copper sulfate solution / cm³	Volume of water / cm³	75	22.5	7.5	Accept 0.0225/2.25 × 10 ⁻² /22 500 and 0.0075/7.5 × 10 ⁻³ /7500
	Concentration of copper sulfate solution / g kg⁻¹		Volume of 100 g kg⁻¹ copper sulfate solution / cm³	Volume of water / cm³					
75	22.5	7.5							
		Ignore units in 2nd row. Do not accept mm ⁻³ /cm ⁻³ /dm ⁻³ /ml							

Question	Marking Guidance	Mark	Comments
08.2	1. Density of 10% protein solution = 1.028; 2. More dense (than 25 g kg⁻¹ copper sulfate solution);	2 (2 x AO2)	

Question	Marking Guidance	Mark	Comments
08.3	16.5 and 22;	1 (AO3)	Must be in correct order

Question	Marking Guidance	Mark	Comments
08.4	Any three from: (Tom) 1. (Healthy donor) not allowed to donate; 2. Less blood collected OR Fewer patients treated; 3. Cause Tom anxiety (about his health); (Lucy) 4. (Gives blood) when it may not be safe (for her) to do so; 5. (Her blood) may not help patients; 6. Her (missed) low haemoglobin goes untreated;	3 max (3 x AO3)	Max 2 for either Tom or Lucy If no credit awarded, max 1 mark for idea of too little haemoglobin left to carry oxygen in blood OR reduced oxygen to respiring tissues